

Student Outcomes Analysis - Complete Code Guide

This guide merges the workflow from the study guide and the case study into one commented Python script. Phases 1-2 (EDA/Data Cleaning) are only needed for the dirty dataset. Phases 3-8 are the required tasks for the clean dataset.

PHASE 1 & 2 (Dirty Dataset Only)

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("student_performance_dataset.csv")

# Explore dataset
print(df.head())
print(df.info())
print(df.describe())

# Check duplicates
print(df["Student_ID"].duplicated().sum())
df.drop_duplicates(subset=["Student_ID"], keep="first", inplace=True)

# Handle missing values (only for dirty dataset)
df.dropna(subset=["Name", "Marks", "Attendance_Percentage"], inplace=True)

# Fix outliers (dirty dataset only)
df.loc[df["Age"] > 100, "Age"] = df["Age"].median()
df.loc[df["Marks"] < 0, "Marks"] = df["Marks"].mean()
df.loc[df["Attendance_Percentage"] > 100, "Attendance_Percentage"] = 100
```

PHASE 3–8 (Required Case Study)

```
# 1. Average Marks
average_marks = df["Marks"].mean()
print(f"Average Marks: {average_marks:.2f}")

# 2. High Performers
high = df[df["Marks"] > 75]
print("Students above 75:", len(high))

# 3. Correlation Analysis
correlation = df["Attendance_Percentage"].corr(df["Marks"])
print(f"Correlation: {correlation:.3f}")
if correlation > 0:
    print("Higher attendance is associated with higher marks.")
elif correlation < 0:
    print("Higher attendance is associated with lower marks.")
else:
    print("No correlation.")

# 4. Feature Engineering
```

```
df["Pass"] = df["Marks"] >= 50

# 5. Pass / Fail Summary
print("Pass:", df["Pass"].sum())
print("Fail:", (~df["Pass"]).sum())

# 6. Histogram
plt.figure(figsize=(8,5))
sns.histplot(df["Marks"], bins=10, kde=True)
plt.title("Distribution of Student Marks")
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.grid(True)
plt.show()
```

Business Summary

- Calculate average marks.
- Identify students scoring above 75.
- Analyze correlation between attendance and marks.
- Create Pass column.
- Count Pass/Fail students.
- Visualize marks distribution.

Recommendations:

- Support struggling students.
- Investigate factors beyond attendance affecting marks.
- Use larger datasets before making policy decisions.